On the Structural Plastica

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August 27, 2026 (revised August 28, 2026)

Abstract

The Structural Plastica Theory (SPT) is based on the concept of spacetime as a deformable, elastic solid called the "structural plastica." By viewing spacetime through this lens, we can reproduce physical phenomena like Newtonian gravity without invoking differential geometry.

The main idea of the plastica is that spacetime itself is a solid sitting within a container space. The two coordinate systems (plastica and container) are related via Lavelle's equation, which converts infinitesimal container distances to infinitesimal plastica distances via the forward Jacobian. From a specific choice of plastica configuration, we'll reconstruct gravity.

Note that while SPT is a theory of gravity, it is not built to hold up to Relativity; instead, it is a "toy model" of reality based on the premise of the plastica.

Section 1: Plastica Kinematics

1.1 Lavelle's Equation

Before we begin, it's important to note that we do **not** have a metric in SPT for raising and lowering indices. Index placement is purely notational for Einstein summation and does not indicate covariance or contravariance.

To start, define the plastica as a map from plastica coordinates to container coordinates:

$$\mathbf{C}(\mathbf{x}) \in \mathbb{R}^4 \qquad \mathbf{C}, \mathbf{x} \in \mathbb{R}^4$$

where $\mathbf{C}$ represents container coordinates and $\mathbf{x}$ represents plastica coordinates. Both are 4-vectors, as SPT's spacetime includes three spatial and one temporal direction.

By the standard identity for coordinate maps:

$$dx_\mu = \frac{\partial x_\mu}{\partial C_i} dC^i$$

Rename the forward Jacobian $\frac{\partial x_\mu}{\partial C_i}$ to $\Lambda_{\mu i}$:

$$dx_\mu = \Lambda_{\mu i} dC^i$$

The above equation will be referred to as Lavelle's equation. Notice the matrix structure of this result; along any axis μ, the distance measure does not depend on dC^μ alone, but instead the sum of contributions from other directions weighted by $\Lambda_{\mu i}$. Dividing through by our time dimension's infinitesimal, dC_4:

$$\frac{dx_\mu}{dC_4} = \dot{x}_\mu = \Lambda_{\mu i}\frac{dC^i}{dC_4} = \Lambda_{\mu i}\dot{C}^i$$

Differentiating once more and using the product rule:

$$\frac{d\dot{x}_\mu}{dC_4} = \ddot{x}_\mu = \frac{d\left(\Lambda_{\mu i}\dot{C}^i\right)}{dC_4} = \frac{d\Lambda_{\mu i}}{dC_4}\dot{C}^i + \Lambda_{\mu i}\frac{d\dot{C}^i}{dx^4} = \frac{d\Lambda_{\mu i}}{dC_4}\dot{C}^i + \Lambda_{\mu i}\ddot{C}^i$$

Chain rule the $d\Lambda_{\mu i}/dC_4$:

$$\frac{d\Lambda_{\mu i}}{dC_4} = \frac{\partial\Lambda_{\mu i}}{\partial C_j}\frac{dC^j}{dC_4} = \frac{\partial\Lambda_{\mu i}}{\partial C_j}\dot{C}^j \implies \ddot{x}_\mu = \frac{\partial\Lambda_{\mu i}}{\partial C_j}\dot{C}^i\dot{C}^j + \Lambda_{\mu i}\ddot{C}^i$$

Chain rule $\dot{C}^i$ and $\dot{C}^j$ to convert them to plastica velocities:

$$\dot{C}^i = \frac{\partial C^i}{\partial x^k}\frac{dx_k}{dC_4} = \frac{\partial C^i}{\partial x^k}\dot{x}_k$$

Rename the backward Jacobian $\frac{\partial C^i}{\partial x^k}$ to Θ^{ik}:

$$\dot{C}^i = \Theta^{ik}\dot{x}_k \implies \ddot{x}_\mu = \frac{\partial\Lambda_{\mu i}}{\partial C_j}\Theta^{ik}\Theta^{jl}\dot{x}_k\dot{x}_l + \Lambda_{\mu i}\ddot{C}^i$$

Note that Θ and Λ are inverses, then use the derivative and a chain rule for $\frac{\partial\Lambda_{ab}}{\partial x^d}$:

$$\Lambda_{ab}\Theta^{bc} = \delta^c_a \implies \frac{\partial}{\partial x^d}(\delta^c_a) = 0 = \frac{\partial}{\partial x^d}\left(\Lambda_{ab}\Theta^{bc}\right) = \frac{\partial\Lambda_{ab}}{\partial x^d}\Theta^{bc} + \Lambda_{ab}\frac{\partial\Theta^{bc}}{\partial x^d}$$

$$\frac{\partial\Lambda_{ab}}{\partial x^d} = \frac{\partial\Lambda_{ab}}{\partial C_e}\frac{\partial C^e}{\partial x^d} = \frac{\partial\Lambda_{ab}}{\partial C_e}\Theta^{ed} \implies 0 = \frac{\partial\Lambda_{ab}}{\partial C_e}\Theta^{ed}\Theta^{bc} + \Lambda_{ab}\frac{\partial\Theta^{bc}}{\partial x^d}$$

$$\implies \frac{\partial\Lambda_{ab}}{\partial C_e}\Theta^{bc}\Theta^{ed} = -\Lambda_{ab}\frac{\partial\Theta^{bc}}{\partial x^d}$$

Plugging into $\ddot{x}_\mu$, with $a \to \mu, b \to i, c \to k, d \to l$:

$$\ddot{x}_\mu = -\Lambda_{\mu i}\frac{\partial\Theta^{ik}}{\partial x^l}\dot{x}_k\dot{x}_l + \Lambda_{\mu i}\ddot{C}^i$$

Rename the curvature tensor $\frac{\partial \Theta^{ik}}{\partial x^l}$ to Ω^{ikl}:

$$\ddot{x}_\mu = -\Lambda_{\mu i}\Omega^{ikl}\dot{x}_k\dot{x}_l + \Lambda_{\mu i}\ddot{C}^i$$

Notice that only one term requires an external container force $\ddot{C}^i$. That term bends incoming forces using the forward Jacobian, same way that velocities are bent. The other term only requires velocity, and appears due to the curvature (not just modification) of the plastica. Dropping the external container force to work with free particles, we get Lavelle deflection:

$$\ddot{x}_\mu = -\Lambda_{\mu i}\Omega^{ikl}\dot{x}_k\dot{x}_l$$

1.2 Perturbations To First Order

Before we move into gravity, it's important that we set the stage first. To do so, we'll be deriving Lavelle deflection for a map given by:

$$C_\mu = x_\mu + \epsilon\alpha_\mu$$

where ϵ is some small number and α_i is the perturbation structure. We'll be tracking our result to first order in ϵ, so anything with a factor ϵ^2 or above will be discarded.

Starting with the backward Jacobian:

$$\Theta_{ij} = \frac{\partial x_i}{\partial x_j} + \epsilon\frac{\partial \alpha_i}{\partial x_j} = \delta_{ij} + \epsilon\frac{\partial \alpha_i}{\partial x_j}$$

which gives us the forward Jacobian

$$\implies \Lambda_{ij} = \Theta^{-1}_{ij} = \delta_{ij} - \epsilon\frac{\partial \alpha_i}{\partial x_j} + O(\epsilon^2)$$

Next, the curvature tensor:

$$\Omega_{ijk} = \frac{\partial \Theta_{ij}}{\partial x_k} = \frac{\partial \delta_{ij}}{\partial x_k} + \epsilon\frac{\partial^2 \alpha_i}{\partial x_j \partial x_k} = \epsilon\frac{\partial^2 \alpha_i}{\partial x_j \partial x_k}$$

Now Lavelle deflection:

$$\ddot{x}_\mu = -\Lambda_{\mu i}\Omega^{ikl}\dot{x}_k\dot{x}_l = -\left(\delta_{\mu i} - \epsilon\frac{\partial \alpha_\mu}{\partial x_i}\right)\left(\epsilon\frac{\partial^2 \alpha^i}{\partial x^k x^l}\right)\dot{x}_k\dot{x}_l$$

$$= \epsilon\frac{\partial^2 \alpha_\mu}{\partial x^k \partial x^l}\dot{x}_k\dot{x}_l$$

Moving the $\partial x^k \partial x^l$ indices down and the $\dot{x}_k\dot{x}_l$ indices up, we get our final result; linearized Lavelle deflection equation:

$$\ddot{x}_\mu = -\epsilon\frac{\partial^2 \alpha_\mu}{\partial x_k \partial x_l}\dot{x}^k\dot{x}^l$$

1.3 Building Gravity

We'll be using the following ansatz for gravity; later, it will be shown why this ansatz was chosen:

$$\alpha_\nu = \frac{\kappa x_4^2 x_\nu}{|\mathbf{r}|^3}$$

where ν runs over spatial indices, $\mathbf{r} = (x_1, x_2, x_3)$, κ is a constant we'll identify later, and α_4 is a function to be introduced later as well. For now, we'll be focusing on the spatial components of this ansatz. Differentiating once for spatial index η:

$$\frac{\partial \alpha_\nu}{\partial x_\eta} = \kappa x_4^2 \frac{\partial}{\partial x_\eta}\left(\frac{x_\nu}{|\mathbf{r}|^3}\right) = \kappa x_4^2 \left(\frac{\partial x_\nu}{\partial x_\eta}\frac{1}{|\mathbf{r}|^3} + x_\nu \frac{\partial}{\partial x_\eta}\left(\frac{1}{|\mathbf{r}|^3}\right)\right)$$

The standard result for $\partial/\partial x_\eta\,(1/|\mathbf{r}|^n)$ is $-nx_\eta/|\mathbf{r}|^{n+2}$, so:

$$= \kappa x_4^2 \left(\frac{\delta_{\eta\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_\eta}{|\mathbf{r}|^5}\right)$$

Differentiating for the $\partial\alpha_\nu/\partial x_4$ case:

$$\frac{\partial \alpha_\nu}{\partial x_4} = \frac{\kappa x_\nu}{|\mathbf{r}|^3}\frac{\partial}{\partial x_4}\left(x_4^2\right) = \frac{2\kappa x_\nu x_4}{|\mathbf{r}|^3}$$

Differentiate once more for spatial indices η and ρ:

$$\frac{\partial^2 \alpha_\nu}{\partial x_\rho \partial x_\eta} = \kappa x_4^2 \frac{\partial}{\partial x_\rho}\left(\frac{\delta_{\eta\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_\eta}{|\mathbf{r}|^5}\right) = \kappa x_4^2 \left(\frac{\partial}{\partial x_\rho}\left(\frac{1}{|\mathbf{r}|^3}\right)\delta_{\eta\nu} - 3\frac{\partial}{\partial x_\rho}\left(\frac{x_\nu x_\eta}{|\mathbf{r}|^5}\right)\right)$$

By the standard result for $1/|\mathbf{r}|^n$, the first term:

$$\frac{\partial}{\partial x_\rho}\left(\frac{1}{|\mathbf{r}|^3}\right) = -\frac{3x_\rho}{|\mathbf{r}|^5}$$

By product rule and the standard result for $1/|\mathbf{r}|^n$, the second term:

$$\frac{\partial}{\partial x_\rho}\left(\frac{x_\nu x_\eta}{|\mathbf{r}|^5}\right) = \frac{\partial x_\nu}{\partial x_\rho} x_\eta \frac{1}{|\mathbf{r}|^5} + x_\nu \frac{\partial x_\eta}{\partial x_\rho}\frac{1}{|\mathbf{r}|^5} + x_\nu x_\eta \frac{\partial}{\partial x_\rho}\left(\frac{1}{|\mathbf{r}|^5}\right)$$

$$= \frac{\delta_{\rho\nu} x_\eta}{|\mathbf{r}|^5} + \frac{\delta_{\eta\rho} x_\nu}{|\mathbf{r}|^5} - \frac{5x_\nu x_\eta x_\rho}{|\mathbf{r}|^7} = \frac{\delta_{\rho\nu} x_\eta + \delta_{\eta\rho} x_\nu}{|\mathbf{r}|^5} - \frac{5x_\nu x_\eta x_\rho}{|\mathbf{r}|^7}$$

Substituting in:

$$\frac{\partial^2 \alpha_\nu}{\partial x_\rho \partial x_\eta} = \kappa x_4^2 \left(-\frac{3x_\rho \delta_{\eta\nu}}{|\mathbf{r}|^5} - \frac{3\delta_{\rho\nu} x_\eta + 3\delta_{\eta\rho} x_\nu}{|\mathbf{r}|^5} + \frac{15x_\nu x_\eta x_\rho}{|\mathbf{r}|^7}\right)$$

$$= \kappa x_4^2 \left(\frac{-3x_\rho \delta_{\eta\nu} - 3\delta_{\rho\nu} x_\eta - 3\delta_{\eta\rho} x_\nu}{|\mathbf{r}|^5} + \frac{15x_\nu x_\eta x_\rho}{|\mathbf{r}|^7}\right)$$

For the $\partial^2\alpha_\nu/\partial x_\rho \partial x_4$ case:

$$\frac{\partial^2\alpha_\nu}{\partial x_\rho \partial x_4} = 2\kappa x_4 \frac{\partial}{\partial x_\rho}\left(\frac{x_\nu}{|\mathbf{r}|^3}\right) = 2\kappa x_4 \left(\frac{\partial x_\nu}{\partial x_\rho}\frac{1}{|\mathbf{r}|^3} + x_\nu \frac{\partial}{\partial x_\rho}\left(\frac{1}{|\mathbf{r}|^3}\right)\right)$$

$$= 2\kappa x_4 \left(\frac{\delta_{\rho\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_\rho}{|\mathbf{r}|^5}\right)$$

by the standard result for $1/|\mathbf{r}|^n$. Note that Clairaut's theorem gives:

$$\frac{\partial^2\alpha_\nu}{\partial x_\rho \partial x_4} = \frac{\partial^2\alpha_\nu}{\partial x_4 \partial x_\rho} = 2\kappa x_4 \left(\frac{\delta_{\rho\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_\rho}{|\mathbf{r}|^5}\right)$$

Finally, in the $\partial^2\alpha_\nu/\partial x_4^2$ case:

$$\frac{\partial^2\alpha_\nu}{\partial^2 x_4} = \frac{2\kappa x_\nu}{|\mathbf{r}|^3}\frac{\partial x_4}{\partial x_4} = \frac{2\kappa x_\nu}{|\mathbf{r}|^3}$$

We can now substitute these blocks into the linearized Lavelle deflection equation:

$$\ddot{x}_\nu = -\epsilon \frac{\partial^2\alpha_\nu}{\partial x_k \partial x_l}\dot{x}^k\dot{x}^l = -\epsilon\left(\frac{\partial^2\alpha_\nu}{\partial x_\rho \partial x_\eta}\dot{x}^\rho\dot{x}^\eta + 2\frac{\partial^2\alpha_\nu}{\partial x_\rho \partial x_4}\dot{x}^\rho\dot{x}^4 + \frac{\partial^2\alpha_\nu}{\partial x_4^2}\dot{x}^4\dot{x}^4\right)$$

$$= -\epsilon\begin{pmatrix} +\kappa x_4^2\left(\frac{-3x_\rho\delta_{\eta\nu} - 3\delta_{\rho\nu}x_\eta - 3\delta_{\eta\rho}x_\nu}{|\mathbf{r}|^5} + \frac{15x_\nu x_\eta x_\rho}{|\mathbf{r}|^7}\right)\dot{x}^\rho\dot{x}^\eta \\ +4\kappa x_4\left(\frac{\delta_{\rho\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_\rho}{|\mathbf{r}|^5}\right)\dot{x}^\rho\dot{x}^4 + \frac{2\kappa x_\nu}{|\mathbf{r}|^3}\dot{x}^4\dot{x}^4 \end{pmatrix}$$

$$= \begin{matrix} -\epsilon\kappa x_4^2\left(\frac{-3x_\rho\dot{x}^\rho\dot{x}_\nu - 3x_\eta\dot{x}^\eta\dot{x}_\nu - 3x_\nu\dot{x}^\eta\dot{x}_\eta}{|\mathbf{r}|^5} + \frac{15x_\nu x_\eta x_\rho\dot{x}^\rho\dot{x}^\eta}{|\mathbf{r}|^7}\right) \\ -4\epsilon\kappa x_4\left(\frac{\dot{x}_\nu\dot{x}_4}{|\mathbf{r}|^3} - \frac{3x_\nu x_\rho\dot{x}^\rho\dot{x}_4}{|\mathbf{r}|^3}\right) - \frac{2\epsilon\kappa x_\nu}{|\mathbf{r}|^3}\dot{x}_4^2 \end{matrix}$$

where I've moved all the fours down (without a metric, as SPT doesn't have one) to make things simpler. Grouping by denominator and cleaning up:

$$\ddot{x}_\nu = -\frac{2\epsilon\kappa x_\nu}{|\mathbf{r}|^3}\dot{x}_4^2 - \frac{4\epsilon\kappa x_4\dot{x}_4}{|\mathbf{r}|^3}\left(\dot{x}_\nu - 3\dot{x}^\rho x_\rho x_\nu\right) + \frac{3\epsilon\kappa x_4^2}{|\mathbf{r}|^5}\left(-\dot{x}^\eta\dot{x}_\eta x_\nu - 2x_\rho\dot{x}^\rho\dot{x}_\nu + \frac{5\dot{x}^\eta x_\eta\dot{x}^\rho x_\rho x_\nu}{|\mathbf{r}|^2}\right)$$

1.4 Analyzing Gravity

The above equation simplifies dramatically for a spatially stationary particle, where $\dot{x}_\nu = 0, \dot{x}_\rho = 0, \dot{x}_\eta = 0$:

$$\ddot{x}_\nu = -\frac{2\epsilon\kappa x_\nu}{|\mathbf{r}|^3}\dot{x}_4^2$$

Rewriting $x_\nu/|\mathbf{r}|$ as $\hat{x}_\nu$:

$$\ddot{x}_\nu = -\frac{2\epsilon\kappa\hat{x}_\nu}{|\mathbf{r}|^2}\dot{x}_4^2$$

Identify $\epsilon = G/c^2$ and $\kappa = M/2$:

$$\ddot{x}_\nu = -\frac{GM\hat{x}_\nu}{|\mathbf{r}|^2}\frac{\dot{x}_4^2}{c^2}$$

Later, we'll find out why this matches Newtonian gravity exactly for a stationary particle.

For a moving particle, corrections are picked up; the first of which are:

$$-\frac{4GMx_4\dot{x}_4\dot{x}_\nu}{2c^2|\mathbf{r}|^3} + \frac{12GMx_4x_\nu\dot{x}_4\dot{x}^\rho x_\rho}{2c^2|\mathbf{r}|^3}$$

The first term scales linearly with velocity and points in the $-\dot{x}_\nu$ direction, providing a drag on velocity. The second term scales with the dot product of $\dot{\mathbf{x}}$ and $\mathbf{x}$ and points in the x_ν direction, creating a radial boost that activates for radial velocity. In the early-time limit (small x_4), these terms vanish as well. In the late-time limit, they dominate. The second set of corrections are:

$$-\frac{3GMx_4^2\dot{x}^\eta\dot{x}_\eta x_\nu}{2c^2|\mathbf{r}|^5} - \frac{3GMx_4^2x_\rho\dot{x}^\rho\dot{x}_\nu}{c^2|\mathbf{r}|^5} + \frac{15GMx_4^2\dot{x}^\eta x_\eta\dot{x}^\rho x_\rho x_\nu}{2c^2|\mathbf{r}|^7}$$

The first term scales with the squared magnitude of velocity, pointing in the x_ν direction, pushing away from the origin. The second term is linear in the dot of position and velocity, so it activates for radial motion and drags velocities. The final term is quadratic in the dot of position and velocity, pushing away from the origin. These all vanish in the early-time limit.

Thus, based on our analysis we can say that Newtonian gravity lives in the early-time and low-velocity limit of the plastica.

A small extra effect of gravity included within SPT is length expansion, which comes from Lavelle's equation:

$$dx_\nu = \Lambda_{\nu i}dC^i = \left(1 - \epsilon\frac{\partial\alpha_\nu}{\partial x_i}\right)dC^i = dC^i\left(1 - \epsilon\kappa x_4^2\left(\frac{\delta_{i\nu}}{|\mathbf{r}|^3} - \frac{3x_\nu x_i}{|\mathbf{r}|^5}\right) - \frac{2\epsilon\kappa x_\nu x_4}{|\mathbf{r}|^3}\right)$$

Set $\epsilon = G/c^2$ and $\kappa = M/2$:

$$= dC^i\left(1 - \frac{GMx_\nu x_4}{c^2|\mathbf{r}|^3} - \frac{GMx_4^2}{2c^2|\mathbf{r}|^3}\left(\delta_{i\nu} - \frac{3x_\nu x_i}{|\mathbf{r}|^2}\right)\right)$$

Lengths contract according to the equation above in the plastica. In the early-time limit, $x_4 > x_4^2$ and the first correction dominates.

1.5 Normal Kinematics

We'll be using a familiar constraint to pin α_4 called the "normal" kinematics (NK) constraint:

$$|\dot{\mathbf{x}}| = c \text{ always}$$

This constraint allows phenomena we see in "normal" space, or a nearly flat plastica. For a stationary particle, $\dot{\mathbf{r}} = \mathbf{0}$, so $\dot{x}_4 = c$. Plugging into our stationary particle Lavelle deflection equation:

$$\ddot{x}_\nu = -\frac{GM\hat{x}_\nu}{|\mathbf{r}|^2}\frac{\dot{x}_4^2}{c^2} = -\frac{GM\hat{x}_\nu}{|\mathbf{r}|^2}\frac{c^2}{c^2} = -\frac{GM\hat{x}_\nu}{|\mathbf{r}|^2}$$

We get Newtonian gravity exactly.

Rewriting this constraint with a dot product and differentiating once:

$$\dot{x}_k\dot{x}^k = c^2 \implies \frac{dc}{dC_4} = 0 = \frac{d}{dC_4}\left(\dot{x}_k\dot{x}^k\right) = \ddot{x}^k\dot{x}_k + \dot{x}^k\ddot{x}_k = 2\ddot{x}^k\dot{x}_k = 0$$

$$\implies \ddot{x}^k\dot{x}_k = 0$$

Acceleration must be purely tangential to the 4-velocity. Splitting into temporal and spatial parts:

$$\ddot{x}^\nu\dot{x}_\nu + \ddot{x}_4\dot{x}_4 = 0 \implies \ddot{x}^\nu\dot{x}_\nu = -\ddot{x}_4\dot{x}_4$$

where ν runs over spatial indices. This pins α_4; the definition is:

$$\frac{\ddot{x}^\nu\dot{x}_\nu}{\dot{x}_4} = \epsilon\frac{\partial^2\alpha_4}{\partial x_k\partial x_l}\dot{x}^k\dot{x}^l$$

Substituting in $\ddot{x}_\nu$ and raising $\dot{x}^\nu$ to match:

$$\epsilon\frac{\partial^2\alpha_4}{\partial x_k\partial x_l}\dot{x}^k\dot{x}^l = \begin{aligned}&-\frac{2\epsilon\kappa x_\nu\dot{x}^\nu}{|\mathbf{r}|^3}\dot{x}_4 - \frac{4\epsilon\kappa x_4\dot{x}^\nu}{|\mathbf{r}|^3}\left(\dot{x}_\nu - 3\dot{x}^\rho x_\rho x_\nu\right)\\ &+\frac{3\epsilon\kappa x_4^2\dot{x}^\nu}{|\mathbf{r}|^5\dot{x}_4}\left(-\dot{x}^\eta\dot{x}_\eta x_\nu - 2x_\rho\dot{x}^\rho\dot{x}_\nu + \frac{5\dot{x}^\eta x_\eta\dot{x}^\rho x_\rho x_\nu}{|\mathbf{r}|^2}\right)\end{aligned}$$

Notice the $\dot{x}_4$ in the denominator of the last term in the LHS, and the cubic in velocity in the same term. This implies that α_4 itself carries a velocity dependent structure. Also notice the signature; the LHS is directly the second derivative, whereas in other components it is the negative second derivative, which is an interesting appearance of a Lorentzian signature from the NK constraint.

From the NK constraint also comes time dilation in the plastica:

$$\dot{x}_k\dot{x}^k = \dot{x}_4^2 + \dot{x}^\nu\dot{x}_\nu = c^2 \implies \dot{x}_4 = \sqrt{c^2 - \dot{x}^\nu\dot{x}_\nu} = c\sqrt{1 - \frac{\dot{x}^\nu\dot{x}_\nu}{c^2}}$$

This is Special Relativity's γ. It's important to note that this result doesn't "fall out" of the theory the same way gravity did; it's a standard consequence of a velocity constraint like NK imposes.

Section 2: Plastica Dynamics

2.1 The Sourcing Equation

To source the plastica, we'll be using a Lagrangian action and the Euler-Lagrange equations. Our action, the Lavelle scalar minus a sourcing tensor, goes as:

$$S = \int \mathcal{L}\, d^3x = \int |\det \Lambda| \Omega^{\nu mn} \Omega_{\nu mn} - \omega^\nu \frac{\partial m}{\partial x_\nu}\, d^3x \qquad \omega^\nu = \frac{\partial^2 C^\nu}{\partial x^k \partial x_k} = \Omega^{\nu k}{}_k$$

where ν runs over spatial indices, m, n run over all indices, and m is our mass density. The first term is the squared norm of the curvature, weighted by volume expansion. The second term is a source tensor T^ν, bound to C_ν itself. Plasticas with just curvature or just stretch will have lower S; plasticas with high curvature and stretching maximize S. We'll refer to the first side of $\mathcal{L}$ as $\mathcal{L}_1$; the second side will be $\mathcal{L}_2$.

The Euler-Lagrange equations give:

$$\frac{\partial \mathcal{L}}{\partial C^\eta} - \frac{\partial}{\partial x_\rho} \frac{\partial \mathcal{L}}{\partial \Theta^{\eta\rho}} + \frac{\partial^2}{\partial x_\sigma \partial x_\rho} \frac{\partial \mathcal{L}}{\partial \Omega^{\eta\rho\sigma}} = 0$$

where η runs over the first three indices. The first term is simple, as no term contains $\mathbf{C}$ directly:

$$\frac{\partial \mathcal{L}}{\partial C^\eta} = 0$$

For the second, we use Jacobi's formula on $|\det \Lambda| = |\det \Theta^{-1}|$:

$$\frac{\partial \mathcal{L}_1}{\partial \Theta^{\eta\rho}} = \frac{\partial}{\partial \Theta^{\eta\rho}} \left(|\det \Lambda|\right) \Omega^{\nu mn} \Omega_{\nu mn} = -|\det \Lambda| \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn}$$

$$\implies -\frac{\partial}{\partial x_\rho} \frac{\partial \mathcal{L}}{\partial \Theta^{\eta\rho}} = \frac{\partial}{\partial x_\rho} \left(|\det \Lambda| \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn}\right)$$

$$= \begin{array}{l} + \dfrac{\partial}{\partial x_\rho} \left(|\det \Lambda|\right) \Lambda^{\eta\rho} \Omega^{\nu mn} \Omega_{\nu mn} + |\det \Lambda| \dfrac{\partial \Lambda^{\eta\rho}}{\partial x_\rho} \Omega^{\nu mn} \Omega_{\nu mn} \\ + |\det \Lambda| \Lambda^{\eta\rho} \dfrac{\partial \Omega^{\nu mn}}{\partial x_\rho} \Omega_{\nu mn} + |\det \Lambda| \Lambda^{\eta\rho} \Omega^{\nu mn} \dfrac{\partial \Omega_{\nu mn}}{\partial x_\rho} \end{array}$$

We'll rename the change-of-curve tensor $\partial \Omega_{\nu mn} / \partial x_\rho = \partial^3 C_\nu / \partial x_m \partial x_n \partial x_\rho$ to $\Phi_{\nu mn\rho}$ and use the chain rule on both $|\det \Lambda|$ and $\Lambda^{\eta\rho}$:

$$\frac{\partial}{\partial x_\rho} \left(|\det \Lambda|\right) = \frac{\partial}{\partial \Theta^{ab}} \left(|\det \Lambda|\right) \frac{\partial \Theta_{ab}}{\partial x_\rho} = -|\det \Lambda| \Lambda^{ab} \Omega_{ab\rho}$$

$$\frac{\partial \Lambda^{\eta\rho}}{\partial x_\rho} = \frac{\partial \Lambda^{\eta\rho}}{\partial \Theta^{cd}} \frac{\partial \Theta_{cd}}{\partial x_\rho} = -\Lambda^{c\eta} \Lambda^{\rho d} \Omega_{cd\rho}$$

$$\implies -\frac{\partial}{\partial x_\rho}\frac{\partial \mathcal{L}_1}{\partial \Theta^{\eta\rho}} = \begin{matrix} + |\det\Lambda|\Lambda^{ab}\Omega_{ab\rho}\Lambda^{\eta\rho}\Omega^{\nu mn}\Omega_{\nu mn} + |\det\Lambda|\Lambda^{c\eta}\Lambda^{\rho d}\Omega_{cd\rho}\Omega^{\nu mn}\Omega_{\nu mn} \\ + |\det\Lambda|\Lambda^{\eta\rho}\frac{\partial \Omega^{\nu mn}}{\partial x_\rho}\Omega_{\nu mn} + |\det\Lambda|\Lambda^{\eta\rho}\Omega^{\nu mn}\frac{\partial \Omega_{\nu mn}}{\partial x_\rho} \end{matrix}$$

Noticing that we can combine the last two terms and using Φ:

$$= \begin{matrix} + |\det\Lambda|\Lambda^{ab}\Omega_{ab\rho}\Lambda^{\eta\rho}\Omega^{\nu mn}\Omega_{\nu mn} + |\det\Lambda|\Lambda^{c\eta}\Lambda^{\rho d}\Omega_{cd\rho}\Omega^{\nu mn}\Omega_{\nu mn} \\ + 2|\det\Lambda|\Lambda^{\eta\rho}\Phi_{\nu mn\rho}\Omega^{\nu mn} \end{matrix}$$

Next, we'll take the third term for the first side of $\mathcal{L}$:

$$\frac{\partial \mathcal{L}_1}{\partial \Omega^{\eta\rho\sigma}} = \frac{\partial}{\partial \Omega^{\eta\rho\sigma}}(\Omega^{\nu mn}\Omega_{\nu mn})|\det\Lambda| = |\det\Lambda|\left(\frac{\partial \Omega^{\nu mn}}{\partial \Omega^{\eta\rho\sigma}}\Omega_{\nu mn} + \Omega^{\nu mn}\frac{\partial \Omega_{\nu mn}}{\partial \Omega^{\eta\rho\sigma}}\right)$$

$$= |\det\Lambda|\left(\delta^{\eta\nu}\delta^{\rho m}\delta^{n\sigma}\Omega_{\nu mn} + \Omega^{\nu mn}\delta^\eta_\nu\delta^\rho_m\delta^\sigma_n\right) = |\det\Lambda|\left(\Omega^{\eta\rho\sigma} + \Omega^{\eta\rho\sigma}\right) = 2|\det\Lambda|\Omega^{\eta\rho\sigma}$$

$$\implies \frac{\partial^2}{\partial x_\sigma \partial x_\rho}\frac{\partial \mathcal{L}_1}{\partial \Omega^{\eta\rho\sigma}} = 2\frac{\partial}{\partial x_\sigma}\left(\frac{\partial}{\partial x_\rho}\left(|\det\Lambda|\Omega^{\eta\rho\sigma}\right)\right)$$

$$= 2\frac{\partial}{\partial x_\sigma}\left(\frac{\partial}{\partial x_\rho}\left(|\det\Lambda|\right)\Omega^{\eta\rho\sigma} + |\det\Lambda|\frac{\partial \Omega^{\eta\rho\sigma}}{\partial x_\rho}\right)$$

By previous results:

$$\frac{\partial}{\partial x_\rho}(|\det\Lambda|) = -|\det\Lambda|\Lambda^{ab}\Omega_{ab\rho}$$

$$\implies \frac{\partial^2}{\partial x_\sigma \partial x_\rho}\frac{\partial \mathcal{L}_1}{\partial \Omega^{\eta\rho\sigma}} = 2\frac{\partial}{\partial x_\sigma}\left(-|\det\Lambda|\Lambda^{ab}\Omega_{ab\rho}\Omega^{\eta\rho\sigma} + |\det\Lambda|\Phi^{\eta\rho\sigma}{}_\rho\right)$$

$$= \begin{matrix} -2\frac{\partial}{\partial x_\sigma}\left(|\det\Lambda|\right)\Lambda^{ab}\Omega_{ab\rho}\Omega^{\eta\rho\sigma} - 2|\det\Lambda|\frac{\partial \Lambda^{ab}}{\partial x_\sigma}\Omega_{ab\rho}\Omega^{\eta\rho\sigma} - 2|\det\Lambda|\Lambda^{ab}\frac{\partial \Omega_{ab\rho}}{\partial x_\sigma}\Omega^{\eta\rho\sigma} \\ -2|\det\Lambda|\Lambda^{ab}\Omega_{ab\rho}\frac{\partial \Omega^{\eta\rho\sigma}}{\partial x_\sigma} + 2\frac{\partial}{\partial x_\sigma}\left(|\det\Lambda|\right)\Phi^{\eta\rho\sigma}{}_\rho + 2|\det\Lambda|\frac{\partial \Phi^{\eta\rho\sigma}{}_\rho}{\partial x_\sigma} \end{matrix}$$

Renaming the curve-of-curve tensor $\Psi^\eta = \partial^4 C^\eta / \partial x^\rho \partial x^\sigma \partial x_\rho \partial x_\sigma$ and expanding derivatives:

$$= \begin{matrix} +2|\det\Lambda|\Lambda^{cd}\Omega_{cd\sigma}\Lambda^{ab}\Omega_{ab\rho}\Omega^{\eta\rho\sigma} + 2|\det\Lambda|\Lambda^{ca}\Lambda^{bd}\Omega_{cd\sigma}\Omega_{ab\rho}\Omega^{\eta\rho\sigma} - 2|\det\Lambda|\Lambda^{ab}\Phi_{ab\rho\sigma}\Omega^{\eta\rho\sigma} \\ -2|\det\Lambda|\Lambda^{ab}\Omega_{ab\rho}\Phi^{\eta\rho\sigma}{}_\sigma - 2|\det\Lambda|\Lambda^{cd}\Omega_{cd\sigma}\Phi^{\eta\rho\sigma}{}_\rho + 2|\det\Lambda|\Psi^\eta \end{matrix}$$

Finally, the third term for the second side of $\mathcal{L}$:

$$\frac{\partial \mathcal{L}_2}{\partial \Omega^{\eta\rho\sigma}} = -\frac{\partial m}{\partial x_\nu}\frac{\partial^2 \omega^\nu}{\partial \Omega^{\eta\rho\sigma}} = -\frac{\partial m}{\partial x_\nu}\frac{\partial^2 \Omega^{\nu k}{}_k}{\partial \Omega^{\eta\rho\sigma}} = -\frac{\partial m}{\partial x_\nu}\delta^{\eta\nu}\delta^{\rho k}\delta^\sigma_k$$

$$\implies \frac{\partial^2}{\partial x_\sigma \partial x_\rho}\frac{\partial \mathcal{L}_2}{\partial \Omega^{\eta\rho\sigma}} = -\frac{\partial^2}{\partial x_\sigma \partial x_\rho}\left(\frac{\partial m}{\partial x_\nu}\delta^{\eta\nu}\delta^{\rho k}\delta^\sigma_k\right) = -\frac{\partial^2}{\partial x_\sigma \partial x_\rho}\left(\frac{\partial m}{\partial x_\nu}\right)\delta^{\eta\nu}\delta^{\rho k}\delta^\sigma_k$$

$$= -\frac{\partial^3 m}{\partial x_\sigma \partial x_\rho \partial x_\nu}\delta^{\eta\nu}\delta^{\rho k}\delta_k^\sigma = -\frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta}$$

Bringing all the terms together:

$$0 = \begin{aligned}&+ |\det\Lambda|\Lambda^{ab}\Omega_{ab\rho}\Lambda^{\eta\rho}\Omega^{\nu mn}\Omega_{\nu mn} + |\det\Lambda|\Lambda^{c\eta}\Lambda^{\rho d}\Omega_{cd\rho}\Omega^{\nu mn}\Omega_{\nu mn} + 2|\det\Lambda|\Lambda^{\eta\rho}\Phi_{\nu mn\rho}\Omega^{\nu mn}\\ &+ 2|\det\Lambda|\Lambda^{cd}\Omega_{cd\sigma}\Lambda^{ab}\Omega_{ab\rho}\Omega^{\eta\rho\sigma} + 2|\det\Lambda|\Lambda^{ca}\Lambda^{bd}\Omega_{cd\sigma}\Omega_{ab\rho}\Omega^{\eta\rho\sigma} - 2|\det\Lambda|\Lambda^{ab}\Phi_{ab\rho\sigma}\Omega^{\eta\rho\sigma}\\ &- 2|\det\Lambda|\Lambda^{ab}\Omega_{ab\rho}\Phi^{\eta\rho\sigma}{}_\sigma - 2|\det\Lambda|\Lambda^{cd}\Omega_{cd\sigma}\Phi^{\eta\rho\sigma}{}_\rho + 2|\det\Lambda|\Psi^\eta\\ &- \frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta}\end{aligned}$$

Factoring, moving T^η, collapsing summations, and using $\mathcal{L}_1$'s definition:

$$\frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta} = \mathcal{L}_1\Omega_{ab\rho}\left(\Lambda^{ab}\Lambda^{\eta\rho} + \Lambda^{a\eta}\Lambda^{\rho b}\right) + 2|\det\Lambda|\begin{pmatrix} + \Lambda^{\eta\rho}\Omega^{\nu mn}\Phi_{\nu mn\rho}\\ + \Lambda^{cd}\Lambda^{ab}\Omega_{cd\sigma}\Omega_{ab\rho}\Omega^{\eta\rho\sigma}\\ + \Lambda^{ca}\Lambda^{bd}\Omega_{cd\sigma}\Omega_{ab\rho}\Omega^{\eta\rho\sigma}\\ - \Lambda^{ab}\Omega^{\eta\rho\sigma}\Phi_{ab\rho\sigma}\\ - \Lambda^{ab}\Omega_{ab\rho}\Phi^{\eta\rho\sigma}{}_\sigma\\ - \Lambda^{cd}\Omega_{cd\sigma}\Phi^{\eta\rho\sigma}{}_\rho \end{pmatrix} + 2|\det\Lambda|\Psi^\eta$$

$$\frac{1}{2}\frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta} = |\det\Lambda|\Psi^\eta + |\det\Lambda|\begin{pmatrix} + \Lambda^{\eta\rho}\Omega^{\nu mn}\Phi_{\nu mn\rho}\\ - \Lambda^{ab}\Omega^{\eta\rho\sigma}\Phi_{ab\rho\sigma} - 2\Lambda^{ab}\Omega_{ab\rho}\Phi^{\eta\rho\sigma}{}_\sigma\\ + \Omega_{cd\sigma}\Omega_{ab\rho}\Omega^{\eta\rho\sigma}\left(\Lambda^{ab}\Lambda^{cd} + \Lambda^{ca}\Lambda^{bd}\right) \end{pmatrix} + \frac{1}{2}\mathcal{L}_1\Omega_{ab\rho}\left(\Lambda^{ab}\Lambda^{\eta\rho} + \Lambda^{a\eta}\Lambda^{\rho b}\right)$$

This is the sourcing equation of the plastica in its full form.

2.2 Analyzing the Sourcing Equation

Before we begin analyzing, let's discuss what adding and subtracting a term means in the sourcing equation. For a fixed LHS, adding a positive term means there is more effect for the same amount of T^η; thus, added terms make the plastica more reactive. Subtracting a term means there is less effect for the same amount of T^η; thus, subtracted terms make the plastica stiffer.

A few semantic terms I will be using should also be discussed beforehand; "bound" quantities are the factors of a product, and thus follow a multiplicative relationship. For a fixed product, one must increase when the other decreases, and both must be nonzero. "Stretch" refers to the physical stretching of the plastica, represented by Λ and Θ.

If we assume the ansatz $C_i = x_i + \epsilon\alpha_i$, we can track the orders of each sourcing equation term as well; we'll do so by assigning each building block an order:

$$\Lambda_{ij} = \delta_{ij} - \epsilon\frac{\partial\alpha_i}{\partial x_j} \sim O(1) \qquad \Omega_{ijk} = \epsilon\frac{\partial^2\alpha_i}{\partial x_j \partial x_k} \sim O(\epsilon) \qquad \mathcal{L}_1 = |\det\Lambda|\Omega^{\nu mn}\Omega_{\nu mn} \sim O(\epsilon^2)$$

Φ and Ψ are $O(\epsilon)$ too, as they're derivatives of Ω.

Starting with the sourcing equation's first term:

$$|\det\Lambda|\Psi^\eta = \left|\det\frac{\partial\mathbf{x}}{\partial\mathbf{C}}\right|\frac{\partial^4 C^\eta}{\partial x^\rho \partial x_\rho \partial x^\nu \partial x_\nu}$$

This term is a Λ-Ψ product, so it's $O(\epsilon)$. It's a biharmonic operator on C^η (double Laplacian) weighted by volume expansion in the plastica. Under the perturbation ansatz, the term goes as:

$$|\det\Lambda|\Psi^\eta = \epsilon\left|1-\epsilon\frac{\partial\alpha_i}{\partial x^i}\right|\frac{\partial^4\alpha^\eta}{\partial x^\rho \partial x_\rho \partial x^\nu \partial x_\nu} = \epsilon\frac{\partial^4\alpha^\eta}{\partial x^\rho \partial x_\rho \partial x^\nu \partial x_\nu} + O(\epsilon^2)$$

Next, the second term:

$$|\det\Lambda|\begin{pmatrix} +\Lambda^{\eta\rho}\Omega^{\nu mn}\Phi_{\nu mn\rho} \\ -\Lambda^{ab}\Omega^{\eta\rho\sigma}\Phi_{ab\rho\sigma} - 2\Lambda^{ab}\Omega_{ab\rho}\Phi^{\eta\rho\sigma}{}_{\sigma} \\ +\Omega_{cd\sigma}\Omega_{ab\rho}\Omega^{\eta\rho\sigma}\left(\Lambda^{ab}\Lambda^{cd} + \Lambda^{ca}\Lambda^{bd}\right)\end{pmatrix}$$

$$= \left|\det\frac{\partial\mathbf{x}}{\partial\mathbf{C}}\right|\begin{pmatrix} +\dfrac{\partial x^\eta}{\partial C_\rho}\dfrac{\partial^2 C^\nu}{\partial x_m \partial x_n}\dfrac{\partial^3 C_\nu}{\partial x^m \partial x^n \partial x_\rho} \\ -\dfrac{\partial x^a}{\partial C_b}\dfrac{\partial^2 C^\eta}{\partial x_\rho \partial x_\sigma}\dfrac{\partial^3 C_a}{\partial x^b \partial x_\rho \partial x_\sigma} - 2\dfrac{\partial x^a}{\partial C_b}\dfrac{\partial^2 C_a}{\partial x^b \partial x_\rho}\dfrac{\partial^3 C^\eta}{\partial x_\rho \partial x_\sigma \partial x^\sigma} \\ +\dfrac{\partial^2 C_c}{\partial x_d \partial x_\sigma}\dfrac{\partial^2 C_a}{\partial x_b \partial x_\rho}\dfrac{\partial^2 C^\eta}{\partial x_\rho \partial x_\sigma}\left(\dfrac{\partial x^a}{\partial C_b}\dfrac{\partial x^c}{\partial C_d} + \dfrac{\partial x^c}{\partial C_a}\dfrac{\partial x^d}{\partial C_b}\right)\end{pmatrix}$$

The first piece is Λ-Λ-Ω-Φ, which is $O(\epsilon^2)$, and binds the plastica's stretch, curvature, and change of curvature together. The second piece is Λ-Ω-Φ, $O(\epsilon^2)$ as well, mirroring the first piece. The third piece is Ω-Ω-Ω-Λ-Λ, $O(\epsilon^3)$, and binds curvature to stretch. At most, this term is cubic in curvature and quadratic in stretch. These terms are already $O(\epsilon^2)$ or $O(\epsilon^3)$, so they have no linearized equivalent.

Finally, the third term:

$$\frac{1}{2}\mathcal{L}_1\Omega_{ab\rho}\left(\Lambda^{ab}\Lambda^{\eta\rho} + \Lambda^{a\eta}\Lambda^{\rho b}\right) \sim O(\epsilon^3)$$

$$= \frac{1}{2}\left|\det\frac{\partial\mathbf{x}}{\partial\mathbf{C}}\right|\frac{\partial^2 C^\nu}{\partial x_m \partial x_n}\frac{\partial^2 C_\nu}{\partial x^m \partial x^n}\frac{\partial^2 C_a}{\partial x_b \partial x_\rho}\left(\frac{\partial x^a}{\partial C_b}\frac{\partial x^\eta}{\partial C_\rho} + \frac{\partial x^a}{\partial C_\rho}\frac{\partial x^\eta}{\partial C_b}\right)$$

This term is $\mathcal{L}$-Ω-Λ-Λ, which is $O(\epsilon^3)$. The Lavelle scalar reappears here, binding volume expansion, curvature, and stretch together in a single term. Because this has no linearized equivalent, the full linearized sourcing equation comes out to:

$$\frac{1}{2}\frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta} = \epsilon\frac{\partial^4\alpha^\eta}{\partial x^\rho \partial x_\rho \partial x^\nu \partial x_\nu} + O(\epsilon^2)$$

2.3 Using the Sourcing Equation

Now that we have both the linearized sourcing equation and the physical choice of α, let's connect the two. First, recall α^η, and note that η only runs over the first three indices:

$$\alpha^\eta = \frac{\kappa x_4^2 x^\eta}{|\mathbf{r}|^3}$$

Now taking the Laplacian:

$$\frac{\partial^2 \alpha^\eta}{\partial x^\rho \partial x_\rho} = \kappa x_4^2 \frac{\partial^2}{\partial x^\rho \partial x_\rho} \left(\frac{x^\eta}{|\mathbf{r}|^3} \right)$$

Note that this is the same as $\nabla^2(\nabla_\eta(1/|\mathbf{r}|))$, so we can rewrite it as $\nabla^\eta \left(\nabla^2(1/|\mathbf{r}|)\right) = \nabla^\eta \left(-4\pi\delta^3(\mathbf{x})\right)$ by the standard result for the Laplacian of $1/|\mathbf{r}|$:

$$\frac{\partial^2 \alpha^\eta}{\partial x^\rho \partial x_\rho} = -4\kappa x_4^2 \pi \frac{\partial \delta^3}{\partial x^\eta}$$

The full biharmonic is then:

$$\frac{\partial^4 \alpha^\eta}{\partial x^\sigma \partial x_\sigma \partial x^\rho \partial x_\rho} = -4\pi\kappa x_4^2 \frac{\partial^3 \delta^3}{\partial x^\eta \partial x^\sigma \partial x_\sigma}$$

Using the linearized source equation:

$$-4\pi\kappa x_4^2 \frac{\partial^3 \delta^3}{\partial x^\eta \partial x^\sigma \partial x_\sigma} = \frac{1}{2\epsilon} \frac{\partial^3 m}{\partial x^k \partial x_k \partial x^\eta}$$

Strip the derivatives:

$$-8\pi\epsilon\kappa x_4^2 \cdot \delta^3(\mathbf{r}) = m$$

A clean point-source. This is why the gravitational ansatz was chosen; choosing $m \sim \kappa\delta_3(\mathbf{x})$ creates it exactly.

Conclusions

Throughout this paper, we've investigated the implications of a plastica-based universe and explored the outcomes thoroughly.

Beginning with space as a deformable solid medium, we derived the conversions of positions, velocities, and accelerations between our two coordinate systems; we then created a gravitational perturbation and constrained masses moving through the plastica to recover Newtonian gravity exactly. Towards the end, we derived the sourcing equation of the plastica from a Lagrangian action and used it to solve for the general sourcing equation for any continuous mass and proved our gravitational ansatz's validity.

There are still, however, some open questions:

- Does the plastica admit gravitational waves?

- Can SPT reproduce General Relativity under the correct choice of action?
- What does SPT look like at second order/higher orders?
- Does light in SPT behave correctly?

These questions are still under research, and their answers yet to be determined.